Term Project: GRPO for Control

Goal. Group Relative Policy Optimization (GRPO) is the reinforcement learning algorithm that transformed the LLM field. However, its potential in robotic system control has thus far remained underexplored. In this project, you will devise a GRPO variant that can successfully solve control tasks. You will demonstrate that this success generalizes across different robotic platforms.

Algorithm design. Take the vanilla GRPO as basis and improve it approach in such a way that it can solve continuous and discrete control tasks with high performance measured as total episode reward. You are free in your design as long as it is a GRPO variant and it works better than PPO and on par with SAC and TD3. Consider complementing group average score with an appropriate value function, potentially in a multi-step TD fashion.

Algorithm analysis. Formally investigate the convergence properties of your algorithm. Your analysis should give a mathematical answer to the questions below:

- Is your algorithm guaranteed to convergence and why?
- Which assumptions should hold for this convergence to be guaranteed?

A mathematical answer only contains symbolic definitions, lemmata, theorems, and corollaries, proofs of all statements and nothing else. **Hint:** Find what the state of the art says about PPO convergence. Then characterize how the GRPO approximation to the advantage compares to the true value function or under which conditions it gives a policy improvement. Use LLMs extensively. Ask them to do the proofs for you. But break their task down to as small and well-defined subtasks as possible, do your own check, and take the next step afterwards.

Evaluation. Evaluate your algorithm as well as SAC, TD3, and the GAE variant of PPO on the following three environments:

- Car Racing from the Gymnasium Farama Box2D suite.
- `cartpole-swingup-v0` and `acrobot-swingup-v0` from DeepMind Control (DMC) suite.

Train your algorithm online and test it on regular evaluation intervals. Plot the learning curve as the observed undiscounted evaluation episode return as a function of the evaluation rounds marked in the X axis as the number of training steps taken before each evaluation. Replicate each experiment on five different seeds. Specify the seed numbers in your report. If a run result is predicted to take unacceptably long, feel free to reduce the maximum number of episodes or the capacities of the used function approximators. Do all your implementations in Python and PyTorch.

Report (70 pts). Use the official NeurIPS template under THIS HYPERLINK for your report. A sufficiently comprehensive report has a roughly 9-page long main content that comprises the following:

- (2.5 pts) An introduction section (1-1.5 pages) that motivates the significance of the studied problem to the reinforcement learning community and the main takeaways of the proposed solution.

- (5 pts) A related work section (that points to the most relevant existing work and which limitation of them the proposed solution improves. This section as a whole will be evaluated as part of the interim report.
- (20 pts) A methodology section that
  - (5 pts) introduces the complete set of basic notation required to describe the key elements of the MDP on which the algorithm is built. This whole criterion will be evaluated as part of the interim report.
  - (7.5 pts) presents the designed algorithm as a pseudocode and describes each of its main components in natural language
  - (7.5 pts) motivates the necessity of each of these components for the solution of the target problem
- (20 pts) A theory section that presents the assumptions, definitions, and statements in purely mathematical terms (15 pts), as well as their purpose and practical implication in natural language (5 pts). The proofs of the statements should be presented in an appendix.
- (20 pts) An experiments section that contains the learning curves of GRPO, PPO, SAC, and TD3 for each of the three environments within a single figure with three subpanels (15 pts) where each model successfully solves the task and their relative performance can be justified by their properties, and a verbal justification of why the results support the main hypothesis of the project (5 pts). 10 points are reserved for the complete results of PPO, SAC, and TD3 to be graded in the interim report.
- (2.5 pts) A conclusion section that clarifies the limitations of the presented work and presents ideas about how this project can be pursued further by the reader.

The points stated within the parentheses can only be earned if the related task is solved successfully. For example, a methodology section lacking logical unity, a theoretical statement lacking a correct proof, or an experiments section reporting the results of an erroneously or incompletely implemented algorithm will receive zero points. Partial task completions will not be remunerated.

Late policy. No late policy is implemented. The system will close immediately at the deadline time that will be set on itsLearning and missing submissions receive zero points.